

Assessing Urban Sprawl and Urban Heat Island Intensity Change in Bangalore, 2000–2025

Introduction

Urbanisation represents one of the most defining trends of the 21st century. Today, 55% of the world's population resides in cities, a figure expected to rise to 70% by 2050 (Habitat, 2021). Urban areas are dynamic places where people, socio-economic activities, and the heart of governance converge.

Accelerated urbanisation has transformed cities into large hubs of manufacturing and consumption, dealing devastating blows to the long-term well-being of both nature and society. There is rapid population growth, urban sprawl that is manifested in informal settlements, worsening traffic congestion, increased air pollution, high energy consumption and adverse climate change impacts and risks such as heat waves, droughts, floods, fires and sea level rise (Aboulnaga et al., 2022). Urban sprawl alters land cover by replacing vegetated surfaces with impervious materials, intensifying surface heating and disrupting local climates. This expansion strengthens Urban Heat Island effects, where newly built-up areas exhibit significantly higher temperatures than their rural or vegetated surroundings.

Research Question

Thus, the following questions arise: How has urban sprawl increased over the decades in urban metropolises? What is the effect of Urban Heat Islands (UHIs)? Is there a correlation between rapid urbanisation and growing temperatures? This study investigates how urban sprawl between 2000 and 2025 has influenced the spatial distribution and intensity of Urban Heat Islands in Bangalore/Bengaluru. It examines whether newly developed built-up areas correspond with increased Land Surface Temperature (LST) and identifies hotspots where urban expansion directly contributes to UHI intensification.

Remote Sensing Application

Remote sensing provides a powerful means of analysing long-term urban development and thermal characteristics, enabling the assessment of Urban Heat Island (UHI) dynamics. This study combines GHSL (Global Human Settlement) built-up layers with Landsat-derived Land Surface Temperature (LST) to investigate how urban sprawl contributes to UHI intensity in Bangalore from 2000 to 2025.

Urban Sprawl Analysis Using GHSL

GHSL built-up grids (Pesaresi et al., 2016) were used to identify the spatial extent of urbanisation at two time periods. These global datasets, derived from Landsat imagery, are particularly suitable for analysing long-term changes in rapidly urbanising regions such as Bangalore. Binary built-up masks were generated for both years, and new built-up areas were extracted by comparing 2025 built-up pixels against 2000. The resulting sprawl map shows extensive development in the eastern and southeastern corridors, aligning with Bangalore's IT-driven growth zones in the last two decades.

LST Derivation and Thermal Remote Sensing

Thermal remote sensing offers an effective way to quantify the surface thermal responses to urbanisation. Landsat 7 ETM+ (2000) and Landsat 8 OLI/TIRS (2025) were used because they provide consistent long-term thermal infrared observations. Brightness Temperature was then calculated using the sensor-specific thermal constants K1 and K2.

To account for surface emissivity variability, NDVI was calculated from the Red and NIR bands. The NDVI-based emissivity method has proven effective in heterogeneous urban environments where vegetation and built-up areas co-exist (Avdan & Jovanovska, 2016). The Proportion of Vegetation (Pv) and emissivity (e) were computed and incorporated into the LST equation. LST values were then converted into Celsius for interpretation.

Urban Heat Island (UHI) Mapping and Change Detection

UHI intensity was derived by reclassifying the continuous LST raster into four data-driven categories: Low, Moderate, High, and Very High UHI. This approach aligns with the conceptual framework of Oke (1982), who connected urban form, reduced vegetation, and increased impervious surfaces to elevated temperatures. UHI maps for 2000 and 2025 reveal temperature gradients across Bangalore, with cooler zones concentrated around parks, lakes, and vegetated pockets, and warmer zones appearing around industrial, commercial, and newly urbanised areas.

A UHI change map was created by subtracting UHI 2000 from UHI 2025, showing areas where heat island intensity increased by 1 to 3 classes. The spatial pattern of UHI

intensification is heterogeneous and reflects the combined influence of urban expansion, land cover modification, and changes in vegetation cover.

Sprawl - UHI Interaction

To examine whether newly built-up areas directly contributed to thermal intensification, the sprawl raster was overlaid with the UHI change raster. This intersection identifies only those pixels where newly urbanised land experienced an increase in UHI intensity. The limited number of overlapping pixels indicates that while urban sprawl contributes to UHI formation, temperature increases also occur in established urban cores due to traffic emissions, industrial heat, and the loss of vegetation.

Answering the Research Question

Through the integration of built-up areas and temperatures, this study demonstrates a clear association between urban expansion and rising UHI intensity in specific parts of Bangalore. Remote sensing data provided the spatial and temporal consistency required to analyse these patterns objectively (Weng, 2009). The findings confirm that rapid urbanisation has contributed to localised pockets of thermal intensification, particularly in the high-growth zones of eastern and southeastern Bangalore.

Methodology

The analysis was done using QGIS, integrating multi-temporal satellite imagery and spatial analysis techniques to investigate the relationship between urban sprawl and UHI intensity in Bangalore between 2000 and 2025. The workflow combines Global Human Settlement Layer (GHSL) built-up datasets (Pesaresi et al., 2016) with thermal infrared measurements derived from Landsat 7 ETM+ and Landsat 8 OLI/TIRS imagery obtained from the United States Geological Survey (USGS, 2019, 2020).

Data Sources

GHSL built-up grids for 2000 and 2025 were used to delineate the spatial extent of urban development. GHSL provides globally consistent 100 m built-up classifications derived from Landsat imagery, suitable for long-term monitoring of urbanisation patterns (Pesaresi et al., 2016). Landsat 7 (2000) and Landsat 8 (2025) Level-1 scenes were acquired from the USGS Earth Explorer platform. Thermal Band 6 (ETM+) and Band 10 (OLI/TIRS) were used for LST retrieval. A Bangalore boundary shapefile and ward polygons supported clipping and zonal analysis.

Data Preparation

All datasets were clipped to the Bangalore boundary. GHSL layers were converted to built-up masks and compared to extract sprawl. Landsat thermal bands were converted to radiance, brightness temperature, and finally LST using NDVI-based emissivity, with results expressed in Celsius

Workflow Summary:

1. Data Collection

- Download GHSL built-up datasets for 2000 and 2025.
- Acquire Landsat 7 ETM+ (2000) and Landsat 8 OLI/TIRS (2025) Level-1 scenes from USGS.
- Collect the Bangalore boundary and ward shapefiles.

2. Pre-processing

- Clip all rasters to the Bangalore boundary.
- Create binary built-up masks (1 = built-up, 0 = non-built-up).
- Extract thermal and multispectral bands from Landsat scenes.

3. Urban Sprawl Mapping

- Identify newly built-up areas by comparing 2000 and 2025 GHSL masks.
- Generate a sprawl raster representing areas converted to built-up between 2000 and 2025.

4. Land Surface Temperature (LST) Derivation

- Convert Landsat thermal DN values to TOA
- Compute Brightness Temperature (BT) using K1 and K2 constants.
- Generate NDVI using Red and NIR bands.
- Estimate vegetation proportion (Pv) and emissivity.
- Derive final LST (°C) for 2000 and 2025.

5. UHI Classification and Change Detection

- Reclassify LST into four UHI intensity categories for each year.
- Calculate the UHI change by subtracting UHI 2000 from UHI 2025.

6. Sprawl–UHI Overlap Analysis

- Overlay the sprawl raster with positive UHI change to identify zones where new built-up areas correspond with UHI intensification.
- Generate a binary overlap map highlighting sprawl-driven UHI hotspots.

7. Mapping

- Produce final maps using QGIS Print Layout (with legend, scale, north arrow)

Table 1. Data Geometry Type.

Data	Geometry	Purpose
GHSL Built-up (2000, 2025)	Raster (100 m) → Binary mask	Identifying built-up areas and detecting urban sprawl
Landsat 7 ETM+ (2000)	Raster (Multiband)	Deriving NDVI, emissivity, and LST for 2000
Landsat 8 OLI/TIRS (2025)	Raster (Multiband)	Deriving NDVI, emissivity, and LST for 2025
Bangalore Boundary	Polygon	Clipping rasters, defining study area
Ward Boundaries	Polygon	Zonal statistics of UHI and LST
Sprawl Mask (2000–2025)	Raster (binary)	Mapping newly built-up areas
UHI Intensity Raster	Raster (categorical)	Identifying thermal hotspots
Sprawl–UHI Overlap	Raster (binary)	Detecting where sprawl triggers UHI rise

Table 2. Data Sources.

Data	Source
GHSL Built-up (2000, 2025)	EC JRC Global Human Settlement Layer: https://ghsl.jrc.ec.europa.eu/
Landsat 7 ETM+ (2000)	USGS Earth Explorer: https://earthexplorer.usgs.gov/
Landsat 8 OLI/TIRS (2025)	USGS Earth Explorer: https://earthexplorer.usgs.gov/
Bangalore Boundary	Public GIS datasets (BBMP) https://github.com/datameet/Municipal_Spatial_Data/tree/master/Bangalore

Final Maps:

LST 2000 and LST 2025 (°C)

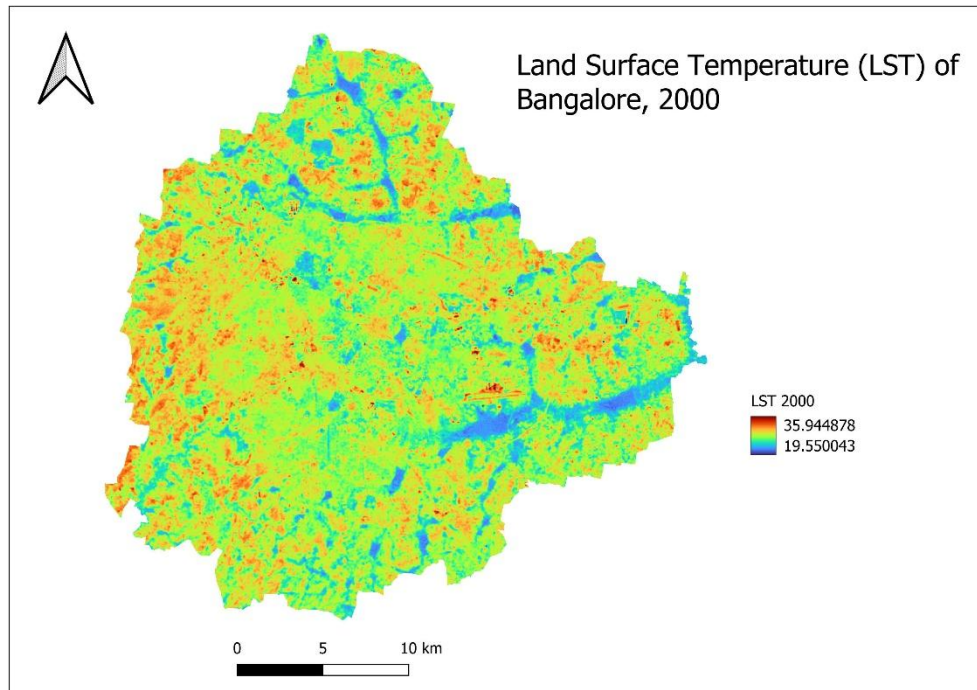


Figure 1. Land Surface Temperature (LST) of Bangalore in 2000.

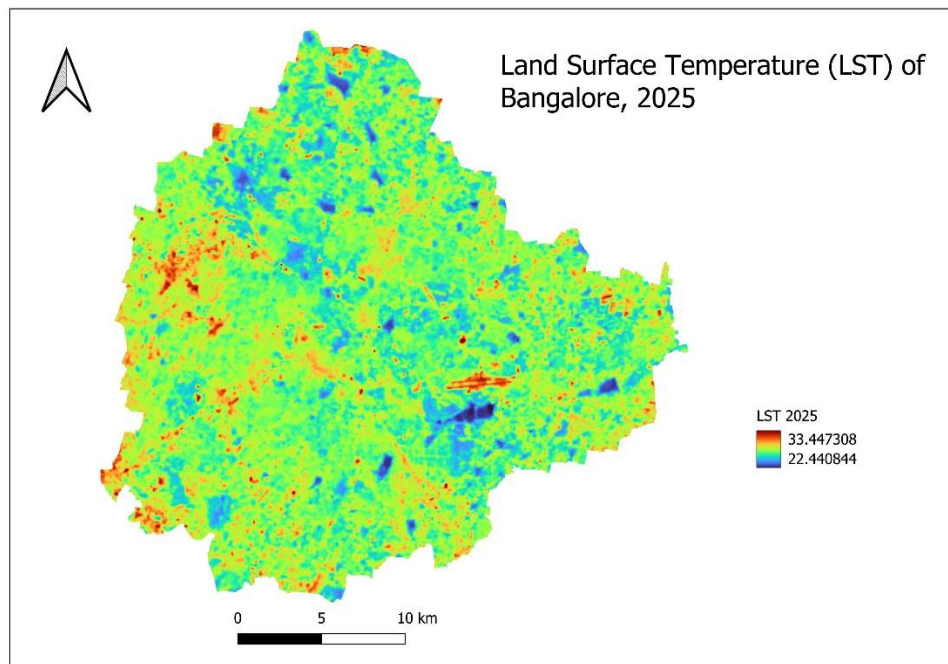


Figure 2. Land Surface Temperature (LST) of Bangalore in 2025.

UHI Reclassified 2000 and UHI Reclassified 2025

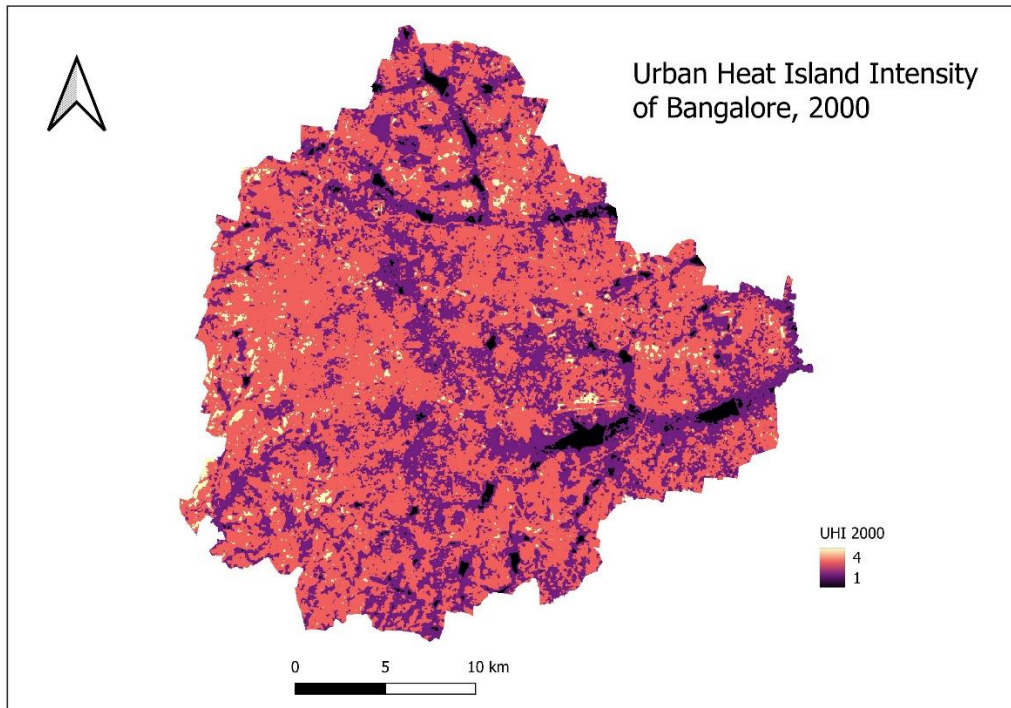


Figure 3. Urban Heat Island (UHI) Intensity Map of Bangalore, 2000.

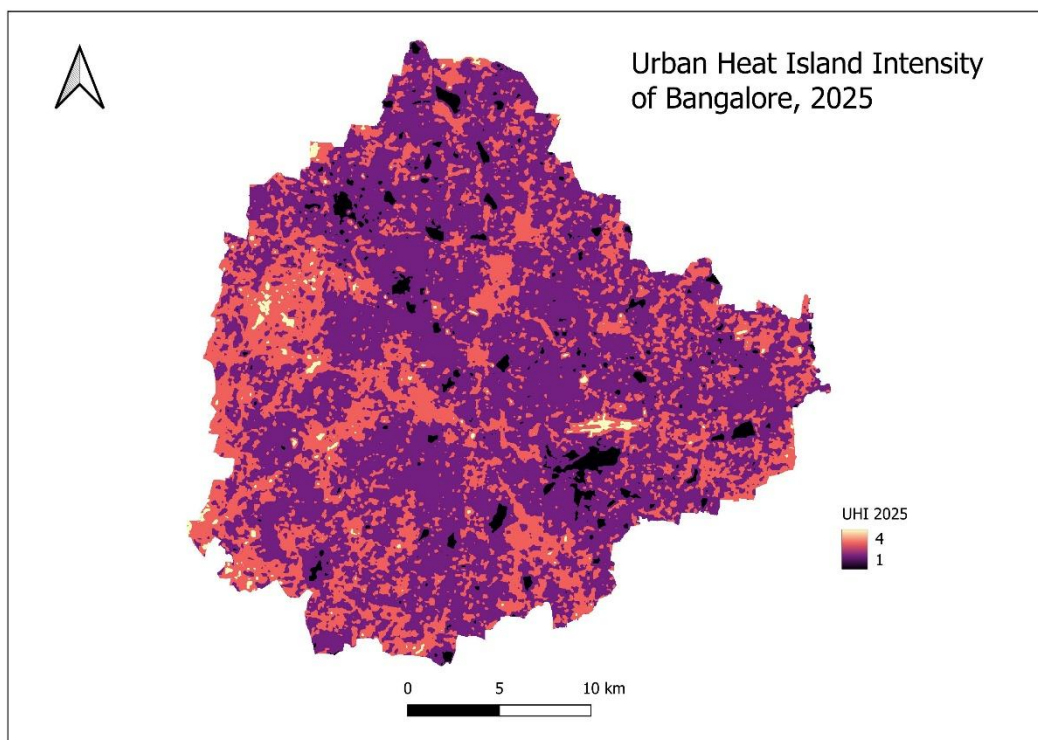


Figure 4. Urban Heat Island (UHI) Intensity Map of Bangalore 2025

UHI Change 2000–2025

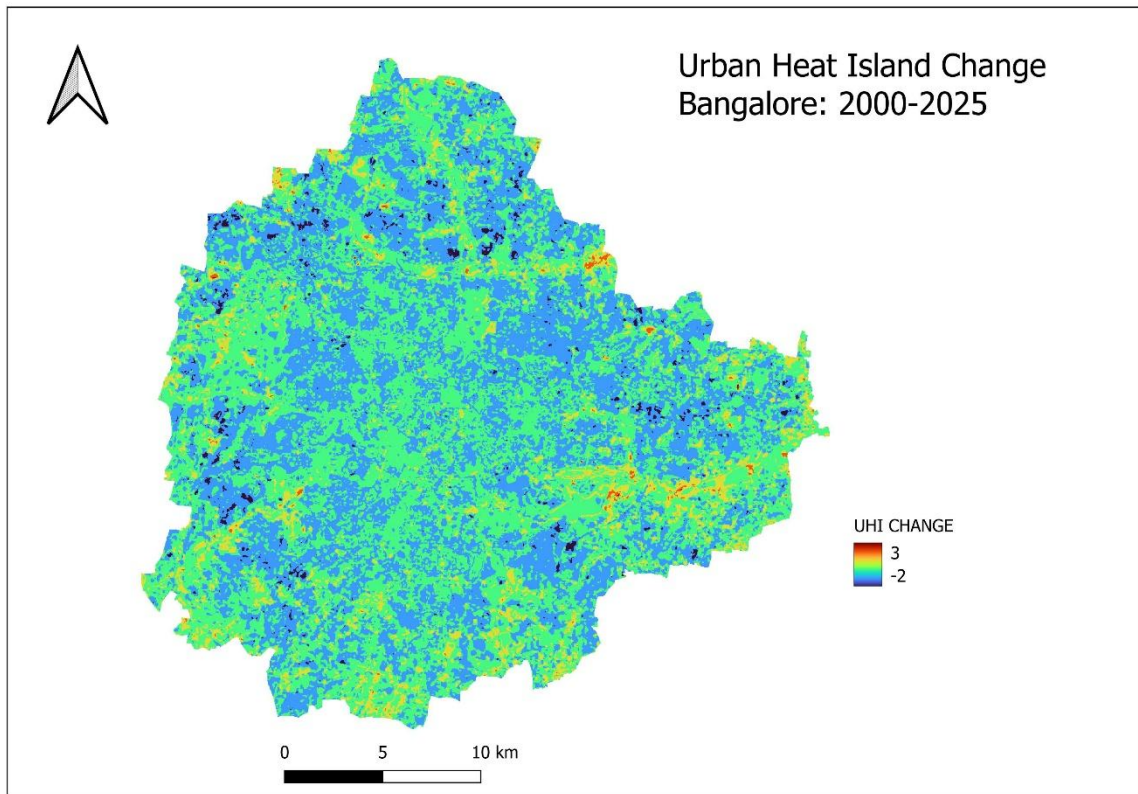


Figure 5. UHI Change Map (2000–2025).

GHSL Built-up 2000 and GHSL Built-up 2025

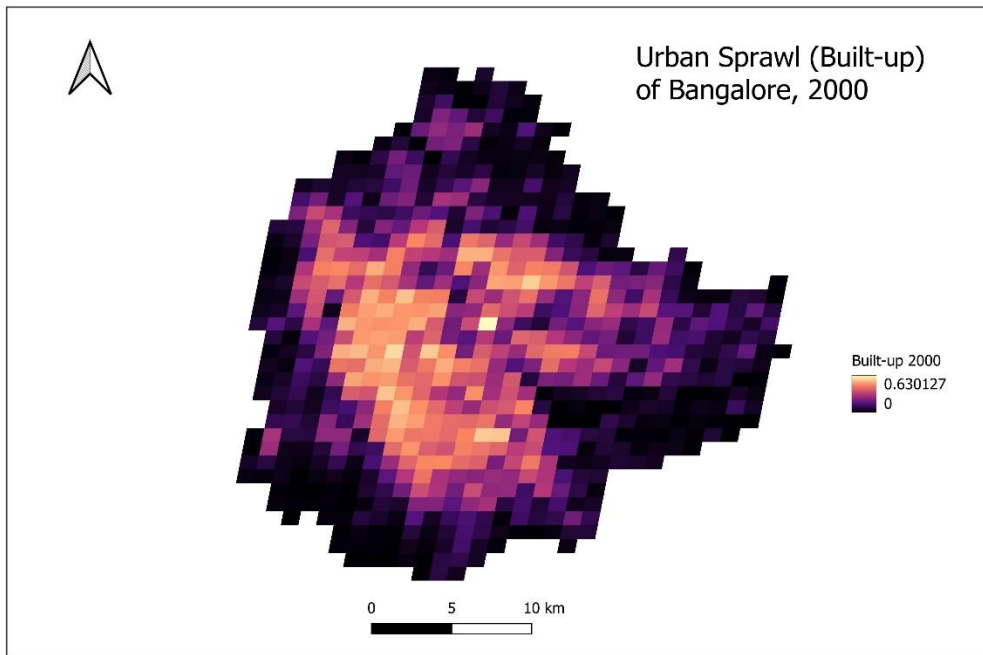


Figure 6. Built-up Areas of Bangalore in 2000 (GHSL).

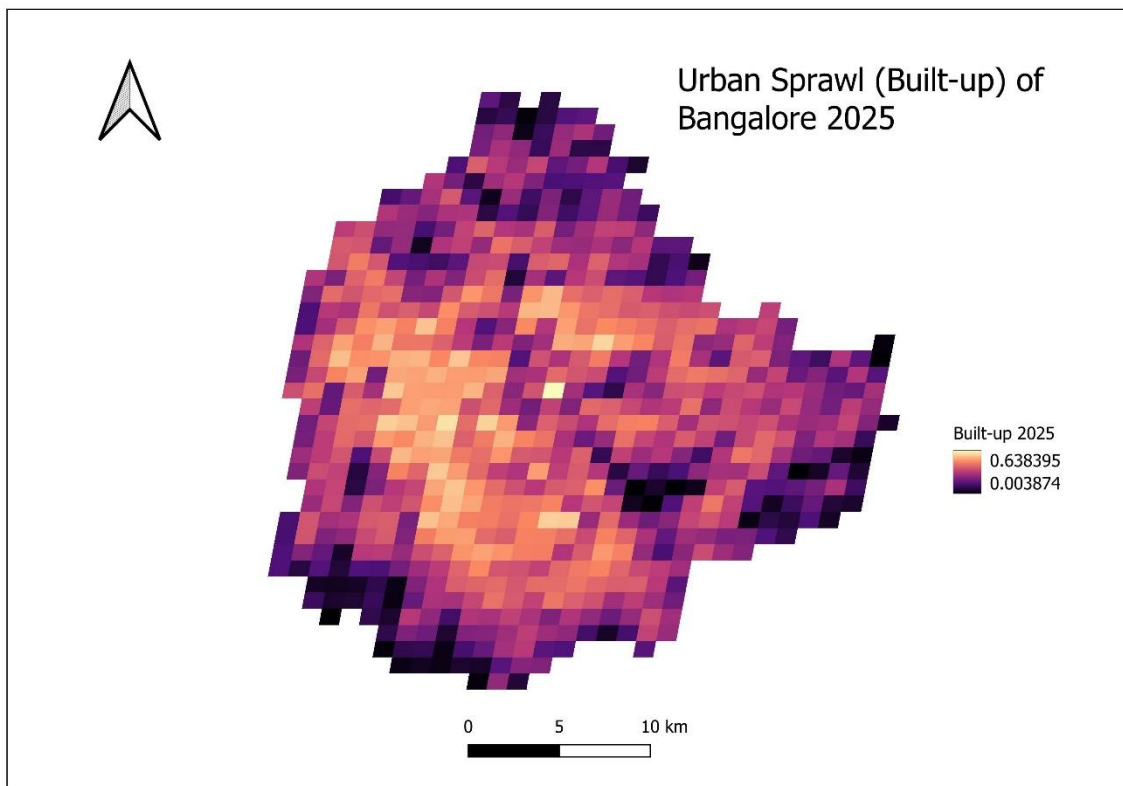


Figure 7. Built-up Areas of Bangalore in 2025 (GHSL).

Sprawl Map (2000–2025)

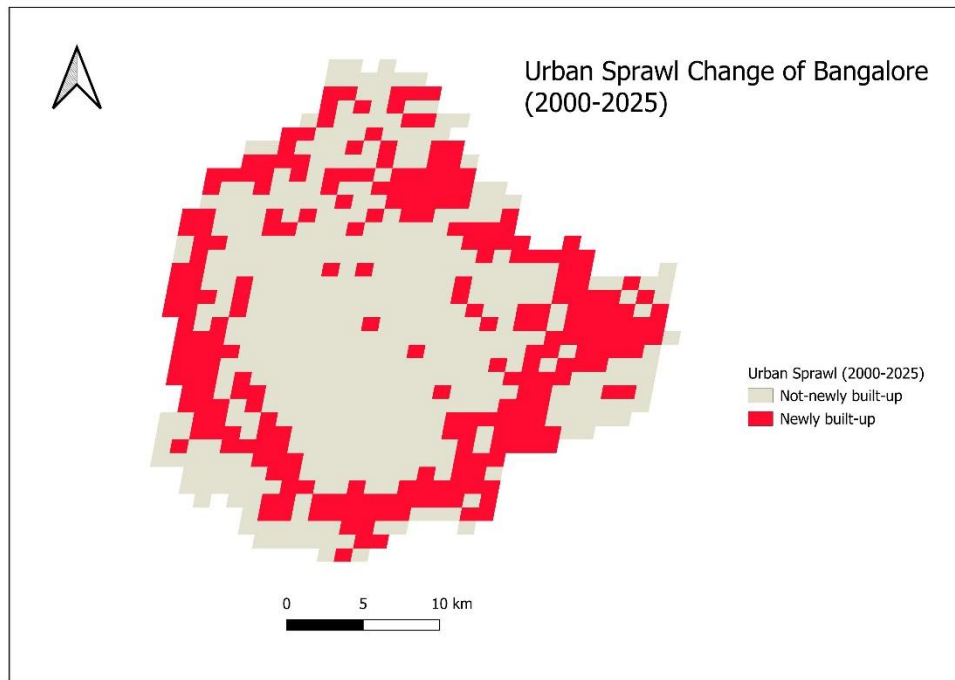


Figure 8. Urban Sprawl Map (2000–2025).

Sprawl-UHI Overlap Map

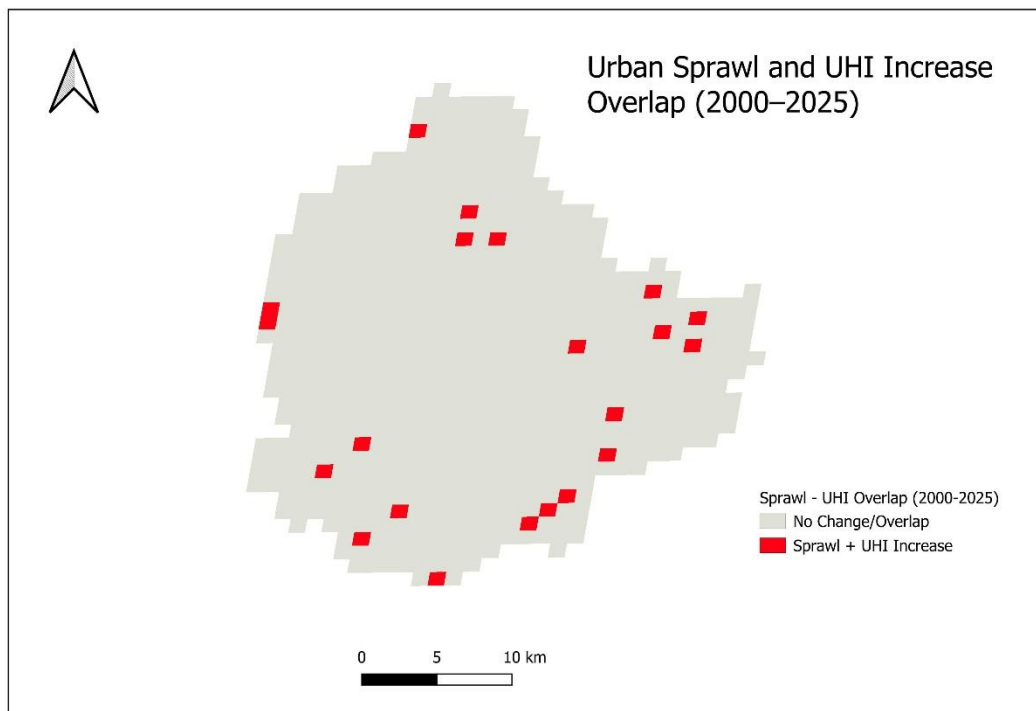


Figure 9. Sprawl–UHI Overlap Map (2000–2025).

Discussion:

The spatial analysis of Bangalore from 2000 to 2025 illustrates how rapid urbanisation has reshaped both the built environment and surface thermal patterns. The Land Surface Temperature (LST) maps show a clear warming trend over 25 years. In 2000, cooler temperatures were concentrated around vegetated pockets, lakes, and peripheral agricultural areas. By 2025, these cooler zones have diminished, and higher LST values have expanded, particularly across the eastern and southeastern growth corridors such as Whitefield, KR Puram, Bellandur, and Sarjapur Road. This pattern reflects the well-documented link between impervious surface expansion and increased surface heating.

The Urban Heat Island (UHI) intensity maps reinforce this trend. In 2000, High and Very High UHI zones were mostly restricted to the central business district and older industrial clusters. In contrast, the 2025 UHI map shows a more dispersed pattern, with newly emerging hotspots in fast-developing peripheral zones. This indicates that Bangalore's UHI structure has transitioned from a core-dominated hotspot to a more polycentric thermal landscape.

The UHI Change Map further highlights areas where UHI intensity increased by one or more classes between 2000 and 2025. While much of the city exhibits little change, several pockets, especially along the expanding eastern and southeastern sectors, show notable warming. A few localised areas exhibit reduced UHI intensity, likely due to increased vegetation or shifts in land use.

The urban sprawl map (2000–2025) clearly identifies broad expansion patterns. New built-up areas are heavily concentrated outside the older city centre, reflecting Bangalore's outward growth driven by IT development and suburban residential demand. Another thing to note is the amount of increase in built-up area over the last 25 years. Built-up area in Bangalore increased dramatically in 2025, showing how rapidly urbanisation has occurred.

Finally, the Sprawl - UHI Overlap Map identifies locations where newly built-up areas directly correspond with increased UHI intensity. Although the overlap pixels are limited, they represent meaningful hotspots where land conversion has actively contributed to surface warming. The relatively small overlap also suggests that UHI intensification is influenced by multiple factors, including traffic density, industrial heat, reduced vegetation, and microclimatic variation and not just urban sprawl alone.

Overall, the combined maps demonstrate that Bangalore's rapid expansion has contributed to the emergence of new thermal hotspots, while also altering the broader spatial pattern of urban temperature distribution.

Conclusion:

This study shows that Bangalore's rapid urban expansion between 2000 and 2025 has contributed to the emergence of new Urban Heat Island hotspots, especially in fast-growing eastern and southeastern corridors. While sprawl plays a clear role in increasing surface temperatures, UHI intensification is also influenced by vegetation loss, traffic density, and changing land-use patterns.

References

- Aboulnaga, M., Ashour, F., Elsharkawy, M., Lucchi, E., Gamal, S., Elmarakby, A., Haggagy, S., Karar, N., Khashaba, N. H., & Abouaiana, A. (2025a). Urbanization and Drivers for Dual Capital City: Assessment of Urban Planning Principles and Indicators for a '15-Minute City.' *Land*, 14(2). Scopus. <https://doi.org/10.3390/land14020382>
- Avdan, U., & Jovanovska, G. (2016). Algorithm for Automated Mapping of Land Surface Temperature Using LANDSAT 8 Satellite Data. *Journal of Sensors*, 2016(1), 1480307. <https://doi.org/10.1155/2016/1480307>
- Imhoff, M. L., Zhang, P., Wolfe, R. E., & Bounoua, L. (2010). Remote sensing of the urban heat island effect across biomes in the continental USA. *Remote Sensing of Environment*, 114(3), 504–513. <https://doi.org/10.1016/j.rse.2009.10.008>
- The energetic basis of the urban heat island—Oke—1982—Quarterly Journal of the Royal Meteorological Society—Wiley Online Library*. (n.d.). Retrieved November 23, 2025, from <https://rmets.onlinelibrary.wiley.com/doi/10.1002/qj.49710845502>
- Pesaresi, M., Schiavina, M., Politis, P., Freire, S., Krasnodębska, K., Uhl, J. H., ... Kemper, T. (2024). [Advances on the Global Human Settlement Layer by joint assessment of Earth Observation and population survey data](#). *International Journal of Digital Earth*, 17(1).
- QGIS Development Team. (2025). *QGIS Geographic Information System* (Version 3.x). QGIS Association. <https://www.qgis.org/qgis.org+1>
- Sustainable Development Goals Cities | *UN-Habitat*. (n.d.). Retrieved May 13, 2025, from <https://unhabitat.org/programme/sustainable-development-goals-cities>
- United Nations Human Settlements Programme. (2021). *World cities report 2020: The value of sustainable urbanization*. UN-Habitat. <https://unhabitat.org/world-cities-report-2020> [qgis.org](https://www.qgis.org)
- United States Geological Survey. (2019). *Landsat 8 data users handbook*. USGS. <https://www.usgs.gov/landsat-missions/landsat-8-data-users-handbook>
- [USGS+2USGS+2](#)

United States Geological Survey. (2024). *Landsat 7 data users handbook*. USGS. <https://www.usgs.gov/landsat-missions/landsat-7-data-users-handbook>
USGS+2USGS+2

Weng, Q. (2009). Thermal infrared remote sensing for urban climate and environmental studies: Methods, applications, and trends. *ISPRS Journal of Photogrammetry and Remote Sensing*, 64(4), 335–344. <https://doi.org/10.1016/j.isprsjprs.2009.03.007>

Submitted by Shravya Omar, MAED 2026
